

Vyomakesh Dundigalla

AI Research Engineer | Agents | Evaluation Infrastructure | Protein Design | GPU Systems

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Summary

AI research engineer focused on LLM agents, post-training alignment, reproducible evaluation systems, protein-design AI, and GPU-level performance optimization. I build agent/eval infrastructure, verifier-backed training loops, trace instrumentation, and practical systems for turning research workflows into reliable experiments.

Research Interests

- Reinforcement learning, pre/post-training, reward modeling, and preference optimization.
- Evaluation frameworks for LLM agents, benchmark design, and instrumentation for failure analysis.
- Bio-AI systems for protein sequence design, structural verification, and lab-feedback loops.

Highlighted Achievements

- Optimized Zerobrew package manager downloads 3.6x faster (6 min to 1.7 min) via racing cancellation and HTTP/2 tuning (Jan 2026).
- Achieved 12.804 microseconds in NVIDIA GPU Mode “nvfp4 block-scaled GEMM” optimization hackathon on B200, targeting speed-of-light performance with CuTeDSL (Dec 2025).
- Built 6 reproducible evaluation environments and benchmarks (2 safety, 2 web, 1 tool-use/MCP, 1 game suite) for PrimeIntellect verifiers (2025).

Experience

LiteFold

Apr 2026 – Present

Founding AI Engineer

Bangalore, India

- Building PLM training and inference pipelines for protein design, from sequence proposal generation to parser-valid outputs and SFT/RLVR loops.
- Building verifier-backed reward infrastructure for PLM workflows, turning model outputs into computable training and evaluation signals.

Zerobrew

Jan 2026

Performance / Systems Engineer (OSS)

Remote

- Delivered 3.6x download optimization (361s to 99s) with racing cancellation and HTTP/2 tuning, eliminating redundant bandwidth usage through atomic coordination (AtomicBool, Notify, and Semaphore).
- Merged to main and immediately impacted all users.

PrimeIntellect-ai (Environments Hub)

2025 – Present

AI Research Engineer (Agents & Evals, OSS)

Remote

- Core contributor to evaluation infrastructure with 5+ merged PRs spanning benchmarks, web agents, and training pipeline integrations.
- Architected 6 evaluation environments across web agents, tool-use systems, and games with a unified API contract for reproducible cross-benchmark testing.
- Standardized reward scoring semantics and trace instrumentation across benchmarks, reducing failure-mode debugging time for research teams.
- Hardened evaluation infrastructure via sandboxed execution boundaries and process isolation for safe, auditable agent runs.
- Key contributions: BALROG game suite (6 simulators), BrowsecompPlus, WebVoyager, JailbreakBench and WMDP safety evals, and HuggingFace MCP integration.

Proposals

VGAC: Verifier-Grounded Agreement-Calibrated RL

2026

Independent research proposal

- Proposes grouped rollouts, verifier signals, and entropy-aware calibration to reduce silent overconfident failures in long-horizon agents.
- Connects training-time reward shaping with inference-time verify/replan routing for agents that need robust uncertainty estimates.

Technical Skills

Languages: Python, C, C++, Rust, Bash, SQL, CUDA
DL / ML: PyTorch, TensorFlow, Keras, NumPy, Matplotlib, Pandas, SciPy
LLM Training & Serving: Transformers, PEFT, vLLM, FSDP, Ray/verl, DeepSpeed ZeRO
GPU: Cutlass, CuTeDSL, FP4/FP8 kernels, throughput and latency optimization
Eval & Agents: Stateful, web, and tool-use benchmarks, reward design, trace instrumentation
Infra & Tooling: Docker, sandboxed execution, API integrations, profiling and performance tooling
Web & Others: HTML, CSS, JavaScript, FastAPI, Node.js, Git, LaTeX

Education

Queen Mary University of London <i>MSc in Business in Blockchain & Society</i>	Sep 2022 – Oct 2023 <i>London, UK</i>
Keshav Memorial Institute of Technology (Aff. JNTU) <i>Bachelor of Technology in Electronics and Communication Engineering</i>	Oct 2017 – Sep 2021 <i>Hyderabad, India</i>

Certifications

Deep Learning Specialization <i>DeepLearning.AI</i>	Apr 2020
Google IT Automation with Python <i>Google</i>	Aug 2020